



THE IMAGINATION UNIVERSITY PROGRAMME

RVfpgaEL2 Lab 5

Creating a Vivado Project

1. Introduction

In order to work with and modify the RVfpgaEL2 System, you will need to build a project that includes all of the Verilog, SystemVerilog, header, configuration, and text files that define the system. In this lab, we show how to create a Vivado project that targets the SoC used in this course to Digilent's Nexys Video Video FPGA board, the -50T version (i.e. RVfpgaEL2-Nexys VideoA7). By following these same steps, you will be able to modify RVfpgaEL2-Nexys VideoA7 and resynthesize it.

IMPORTANT: If you haven't already done so, install Xilinx's Vivado 2022.2 as explained in the Getting Started Guide.

IMPORTANT: The core frequency of the default RVfpgaEL2 System for a Nexys Video Video board is 12.5 Mhz (bitstream provided at **[RVfpgaEL2Nexys VideoNoDDRPath]/src/rvfpgaNexys Video.bit**). A core with a frequency of 25 MHz also works in all our exercises, but it provides some timing violations when the bitstream is created in Vivado, thus we do not use it as the default frequency. The specific configuration used for the core is specified in Lab 11.

2. Creating a Vivado Project for RVfpgaEL2-Nexys VideoA7

You will use Xilinx's Vivado Design Suite to build the RVfpgaEL2-Nexys VideoA7 system using the RTL, the Verilog files that define the system. Follow these steps, detailed below, to build the RVfpgaEL2-Nexys VideoA7 system and target it to a Nexys Video Video FPGA board.

Step 1. Open Vivado

Step 2. Create a new RTL project

Step 3. Add the RTL source files and the constraint files

Step 4. Select Nexys Video Video board's FPGA as the target

Step 5. Set rvfpgaNexys Video as the Top Module, set *common_defines.vh* as global, set *el2_pdef* both as global and as a System Verilog file, add *boot_main.mem* to the project, include folders, and add *-sweep* option

Step 6. Generate bitstream

Step 1. Open Vivado

If you did not install Vivado on your machine as described in the RVfpga Getting Started Guide, do so now. Be sure to install the board files as well.

Now, run Vivado (in **Linux**, open a terminal and type: `vivado`; in **Windows**, open Vivado from the Start menu). The Vivado welcome screen will open. Click on Create Project (see Figure 1).

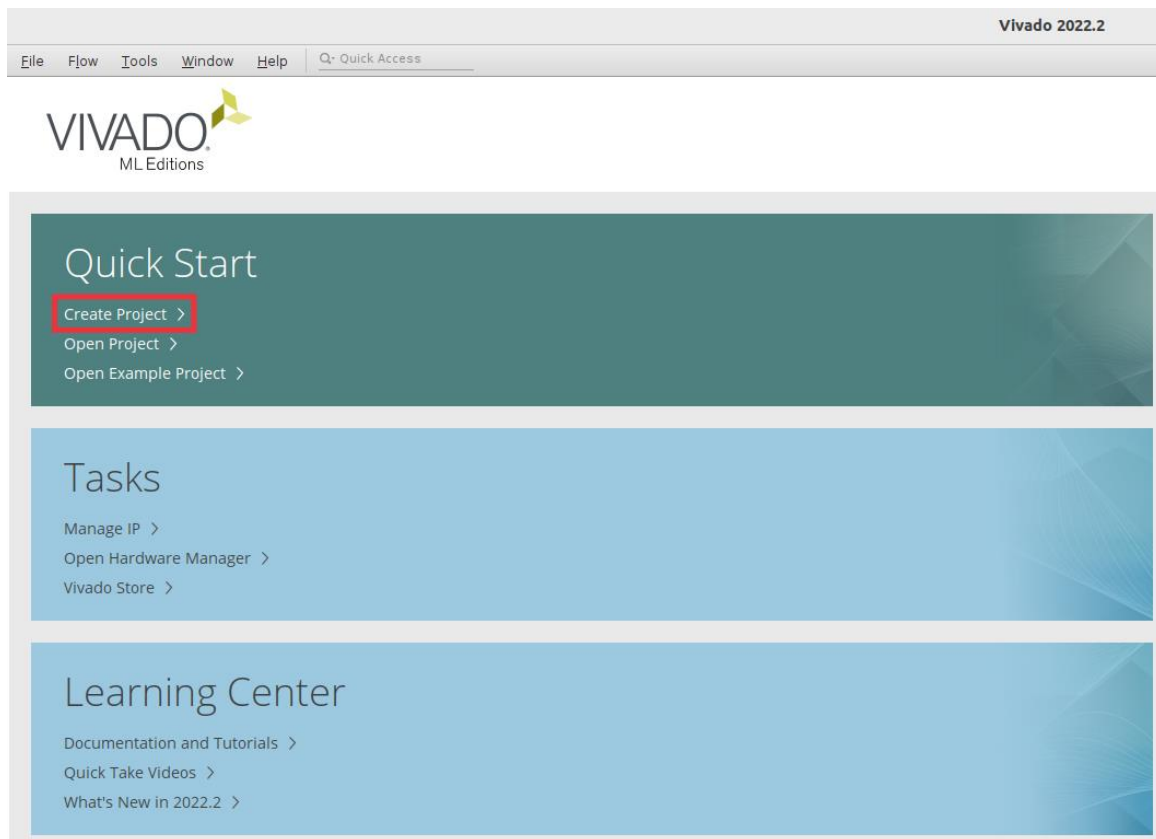


Figure 1. Vivado welcome screen: Create Project

Step 2. Create a new RTL project

The Create a New Vivado Project Wizard will now open (see Figure 2). Click Next.

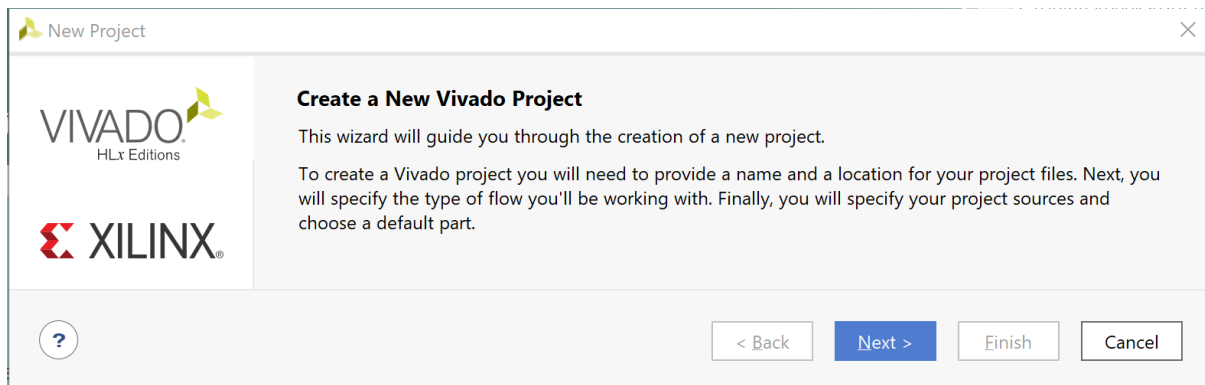
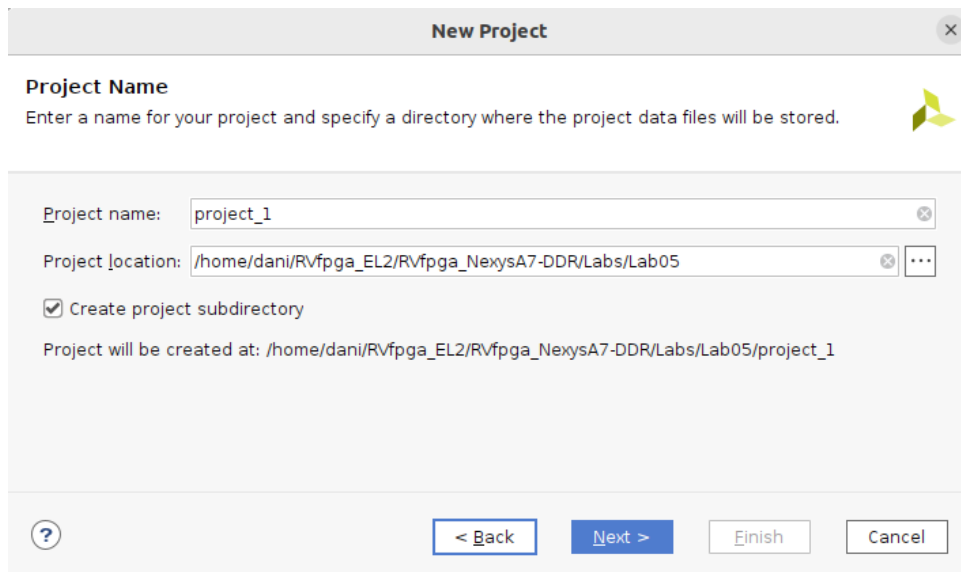


Figure 2. Create a New Vivado Project Wizard

Give a name to your project and place it in the `[RVfpgaEL2Nexys VideoNoDDRPath]/Labs/Lab05` folder. Select option *Create project subdirectory*. Then click Next (see Figure 3).



New Project

Project Name
Enter a name for your project and specify a directory where the project data files will be stored.

Project name:

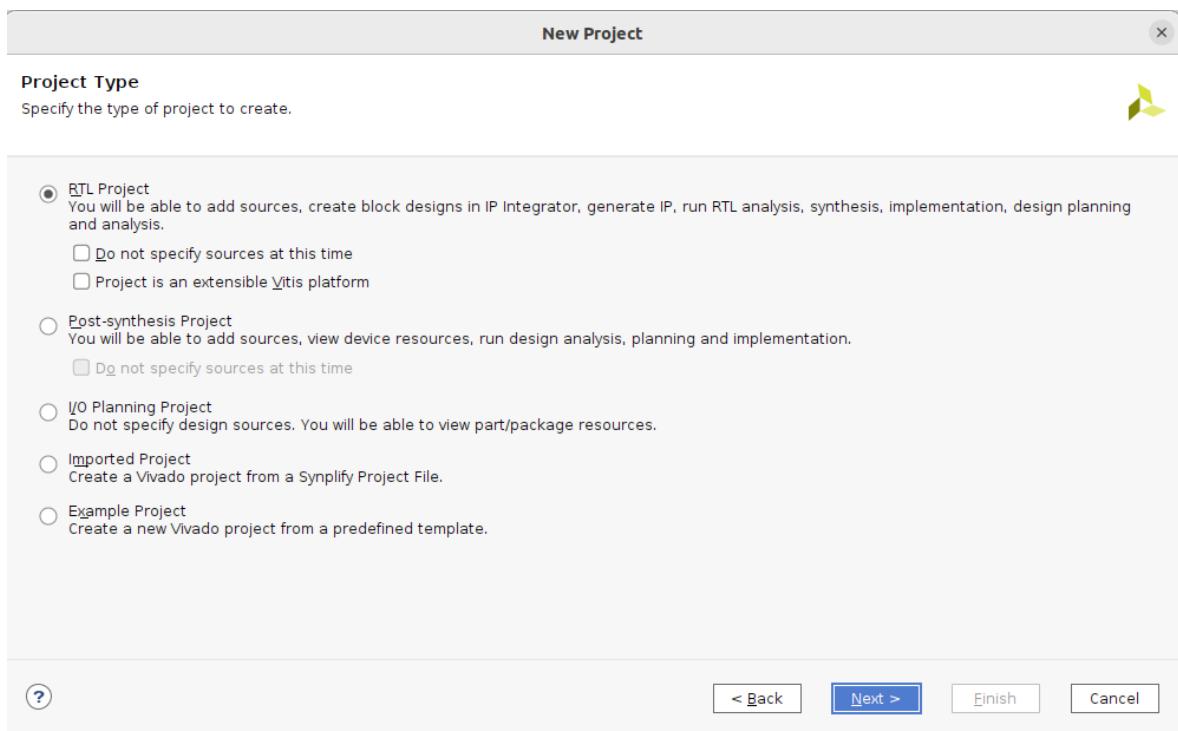
Project location:

☒ Create project subdirectory

Project will be created at: /home/dani/RVfpga_EL2/RVfpga_NexysA7-DDR/Labs/Lab05/project_1

Figure 3. Project Name

Select the project type as RTL Project, and click Next (see Figure 4).



New Project

Project Type
Specify the type of project to create.

☒ **RTL Project**
You will be able to add sources, create block designs in IP Integrator, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.
☐ Do not specify sources at this time
☐ Project is an extensible Vitis platform

☐ **Post-synthesis Project**
You will be able to add sources, view device resources, run design analysis, planning and implementation.
☐ Do not specify sources at this time

☐ **I/O Planning Project**
Do not specify design sources. You will be able to view part/package resources.

☐ **Imported Project**
Create a Vivado project from a Synplify Project File.

☐ **Example Project**
Create a new Vivado project from a predefined template.

Figure 4. RTL Project

Step 3. Add the RTL source files and the constraint files

In the Add Sources window, click on Add Directories, and select *[RVfpgaEL2NexysVideoNoDDRPath]/src* (see Figure 5). Make sure both of the following options are selected (as shown in Figure 5):

- Scan and add RTL include files into project
- Add sources from subdirectories

Then click Next.

New Project ✕

Add Sources
Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.

	Index	Name	Library	HDL Source For	Location
+	1	src	xil_defaultlib	Synthesis & Simulation ▾	/home/dani/Rvfpga_EL2/Rvfpga_NexysA7-DDR

Add Files
Add Directories
Create File

☒ Scan and add RTL include files into project
☐ Copy sources into project
☒ Add sources from subdirectories

Target language: Verilog ▾
Simulator language: Mixed ▾

< Back
Next >
Finish
Cancel

Figure 5. Add Sources

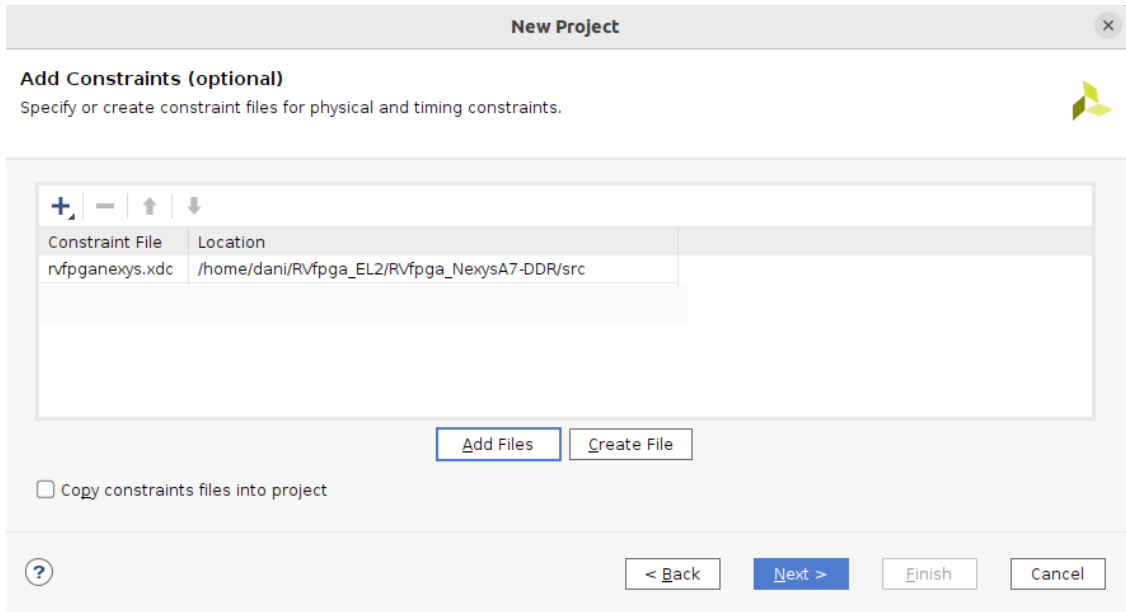
You will now add the constraints for the system. These files map the signal names to the pins on the board. For example, the LEDs on the Nexys Video Video FPGA board are connected to FPGA pins on the board through traces in the PCB. Vivado must know this so that it maps the correct signal name in the RTL to the correct FPGA pin.

Note that the signal name `o_led` is the name used in the Verilog code to drive the Nexys Video Video board's LEDs.

In the Add Constraints window, click on Add Files and select the following files (see Figure 6):

[RVfpgaEL2Nexys VideoNoDDRPath]/src/rvfpgaNexys Video.xdc

Then click Next.



New Project

Add Constraints (optional)
Specify or create constraint files for physical and timing constraints.

Constraint File	Location
rvfpganexys.xdc	/home/dani/RVfpga_EL2/RVfpga_NexysA7-DDR/src

☐ Copy constraints files into project

Figure 6. Add Constraints

Step 4. Select Nexys Video Video board's FPGA as the target

In the Default Part window, select xc7a200tsbg484-1 (see Figure 7). You may use the Search box to narrow down the results.

Click Next.

Search: (1 match)

Part	I/O Pin Count	Available IOBs	LUT Elements	FlipFlops	Block RAMs
xc7a200tsbg484-1	484	285	134600	269200	365

Figure 7. Select target part

In the New Project Summary window, click Finish (see Figure 8).

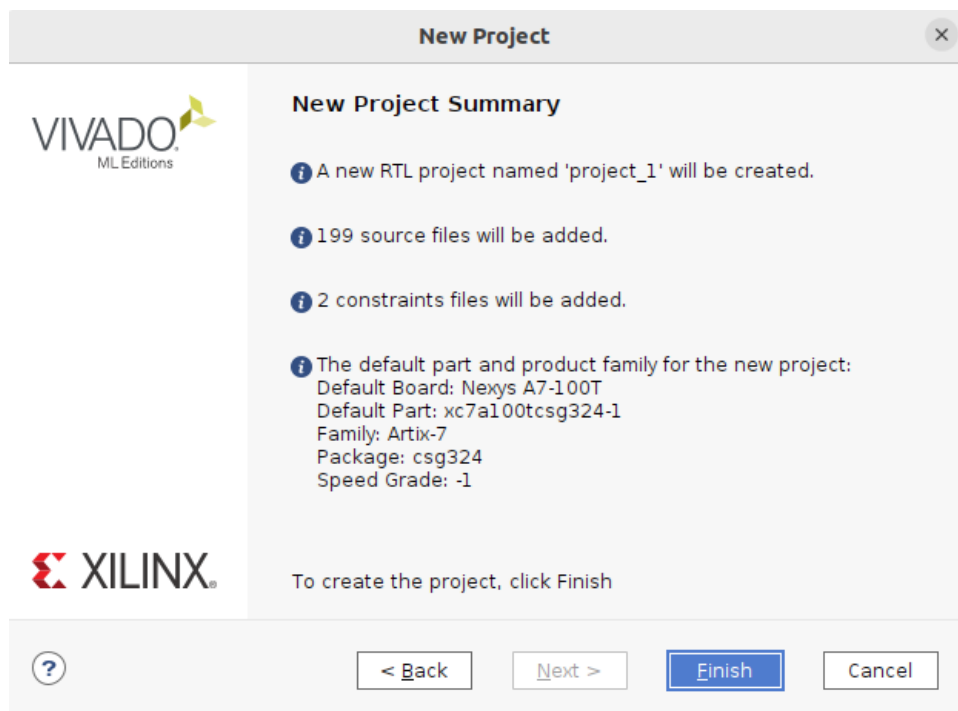


Figure 8. New Project Summary Window

Step 5. Set rvfpgaNexys Video as Top Module, set common_defines.vh as global, set el2_pdef as global and System Verilog, add *boot_main.mem* to the project, include folders, and add *-sweep* option.

Set rvfpgaNexys Video as the Top Module: In the Sources pane, scroll down under Design Sources, right-click on the *rvfpganexys* module, and select Set as Top (see Figure 9). (You can also find the *rvfpganexys* module by typing this name in the search box, as shown.) This sets *rvfpganexys* as the highest-level module in the hierarchy and the target to be synthesized and implemented onto the FPGA. After setting *rvfpganexys* as the top module, the hierarchy will update.

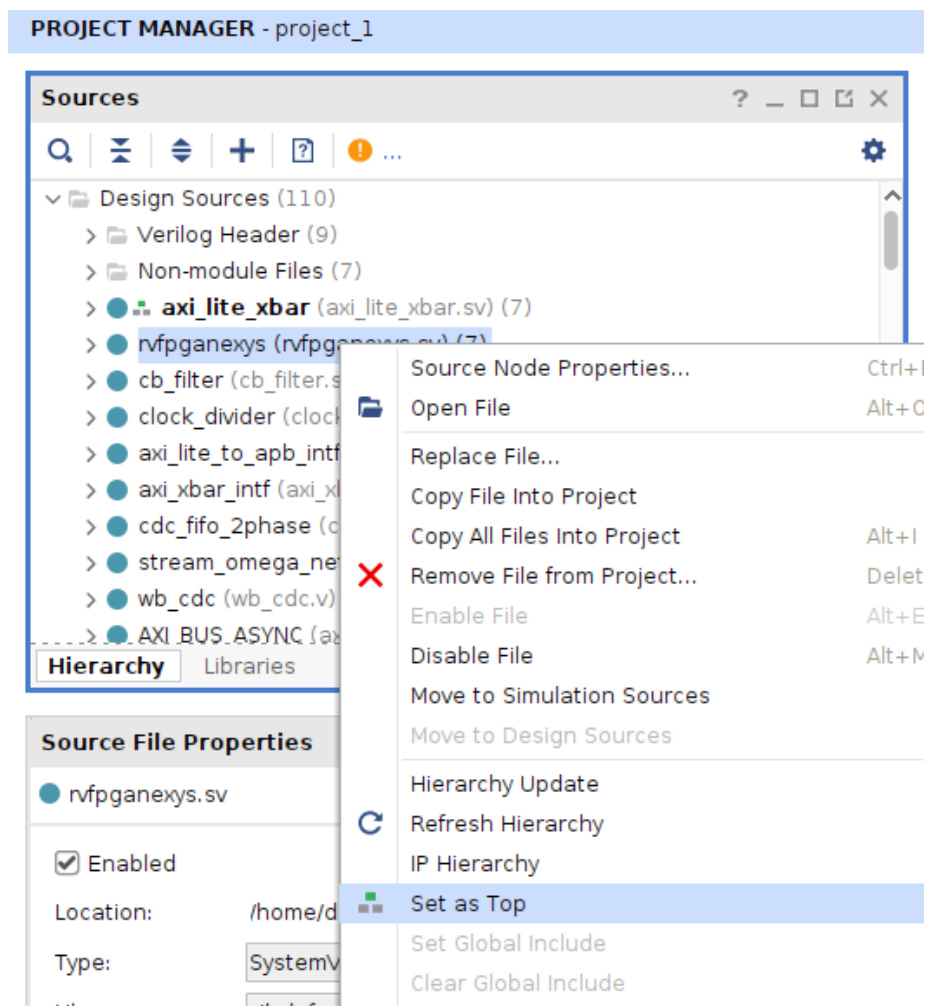


Figure 9. Set rvfpgaNexys Video as top module

Set files *common_defines.vh* and *el2_pdef* as Global Include and *el2_pdef* as SystemVerilog: Now, still in the Sources pane under Design Sources, expand the Non-modules file group and click on *common_defines.vh*. The properties of the file will then open in the Source File Properties pane, just below the Sources pane. Click on Global Include to tick that box (see **Figure 10**). The hierarchy will now update and include that file in Design Sources/Global Include.

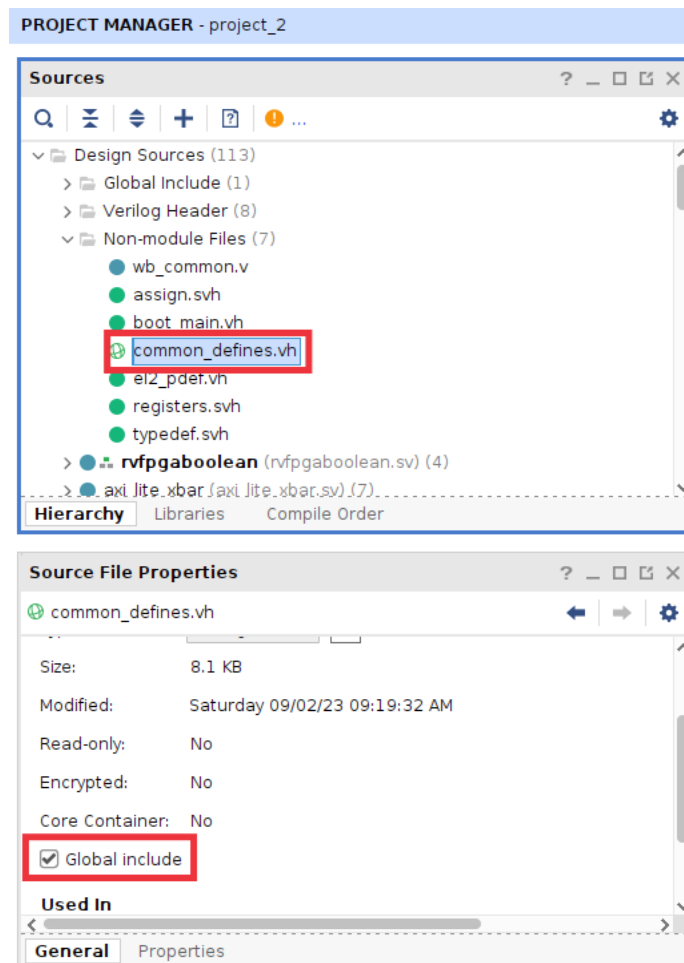


Figure 10. Set *common_defines.vh* as a Global include file

Do the same with file *el2_pdef*. Then, on the same Window (Source File Properties), set the type of this file as SystemVerilog (see Figure 11).

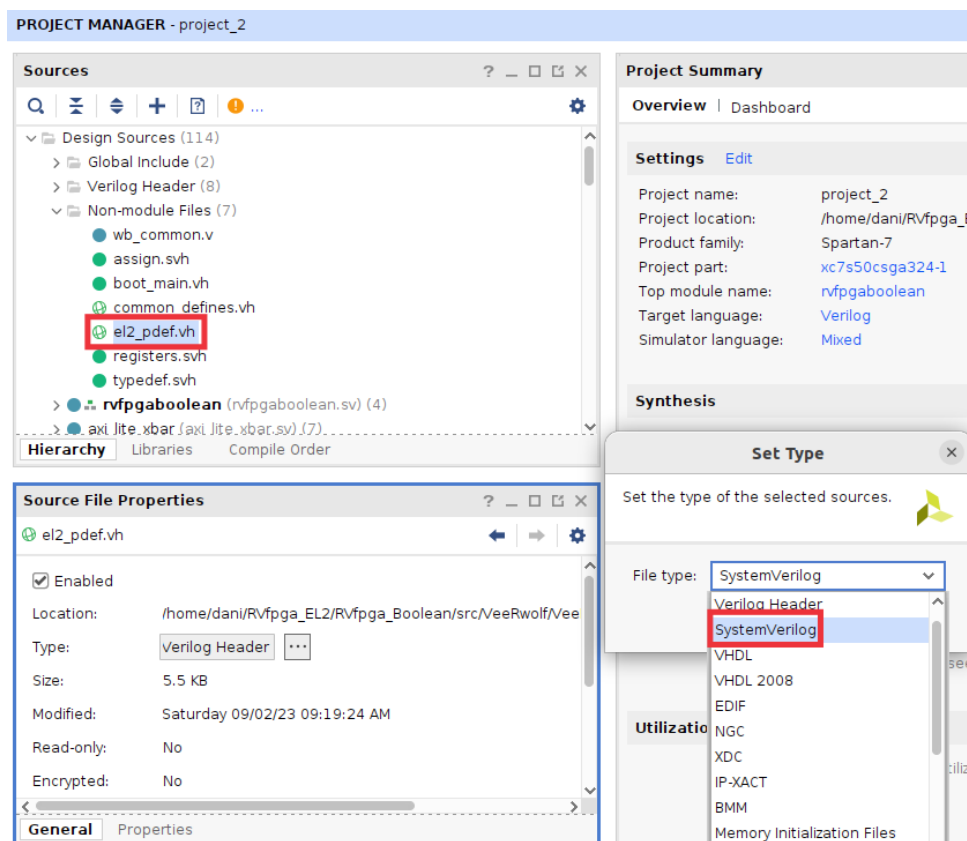


Figure 11. Set the type of *el2_pdf.vh* as SystemVerilog

Add boot_main.mem to the project: In the Flow Navigator pane, click on Add Sources, leave the default option (“Add or create design sources”), and click on Add Files (see Figure 12). Navigate to *[RVfpgaEL2Nexys VideoNoDDRPath]/src/VeeRwolf/BootROM/sw* and select *boot_main.mem* (as shown in Figure 12). The hierarchy will update and include that file in Design Sources/Memory File.

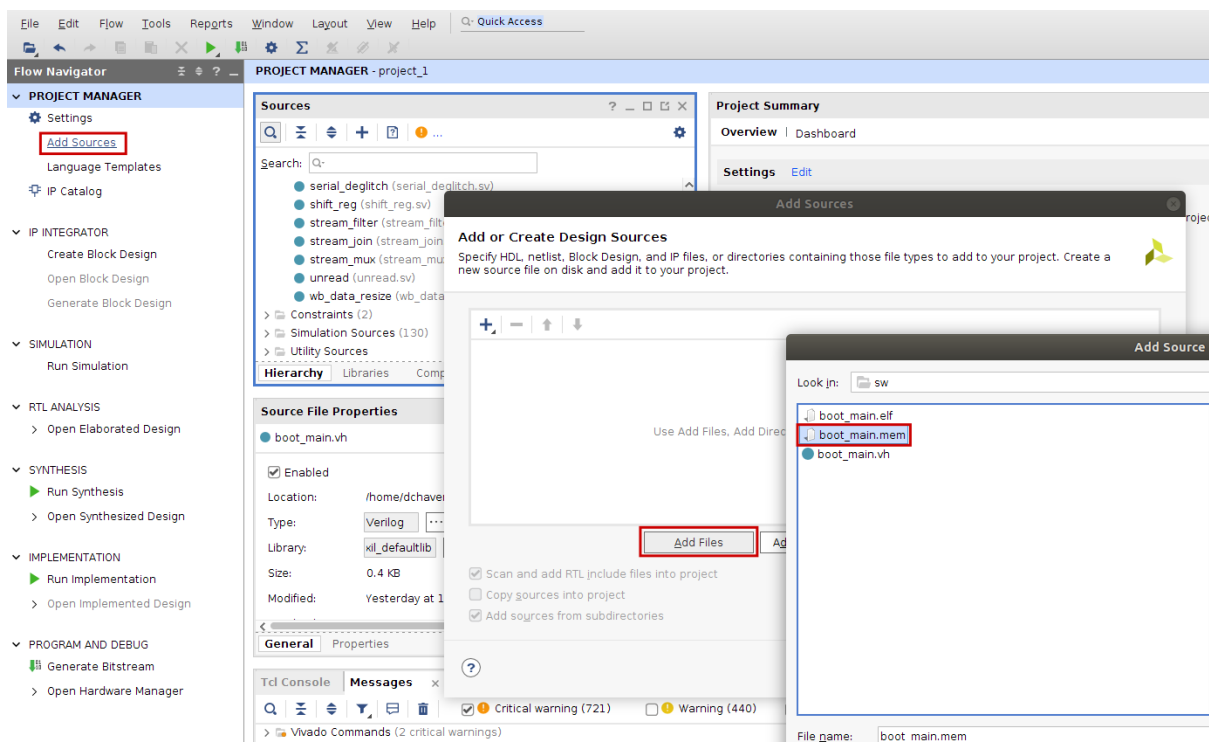


Figure 12. Add Memory File *boot_main.mem*

This file (*boot_main.mem*) is used for initializing the Boot ROM of our SoC by invoking it as a parameter in file *[RVfpgaEL2Nexys VideoNoDDRPath]/src/rvfpgaNexys Video.sv*:

```
`default_nettype none
module rvfpganexys
    #(parameter bootrom_file = "boot_main.mem")
    (input wire      clk,
     input wire      rstn,
```

The Getting Started Guide contains more information about this file.

Include folders: Include two folders for the Pulp Platform (see Figure 13), which we use for the AXI Interconnect and for some generic modules in our SoC (see <https://pulp-platform.org/>). In the Flow Navigator pane, click on *Settings*, and on the window that opens

click on *General* and then on *Verilog options* (). In the new window, add the two following include directories by clicking on  and browsing to the directories:

```
[RVfpgaEL2Nexys VideoNoDDRPath]/src/VeeRwolf/Interconnect/AxiInterconnect/pulp-
platform.org__axi_0.25.0/include
[RVfpgaEL2Nexys VideoNoDDRPath]/src/OtherSources/pulp-platform.org__common_cells_1.20.0/include
```

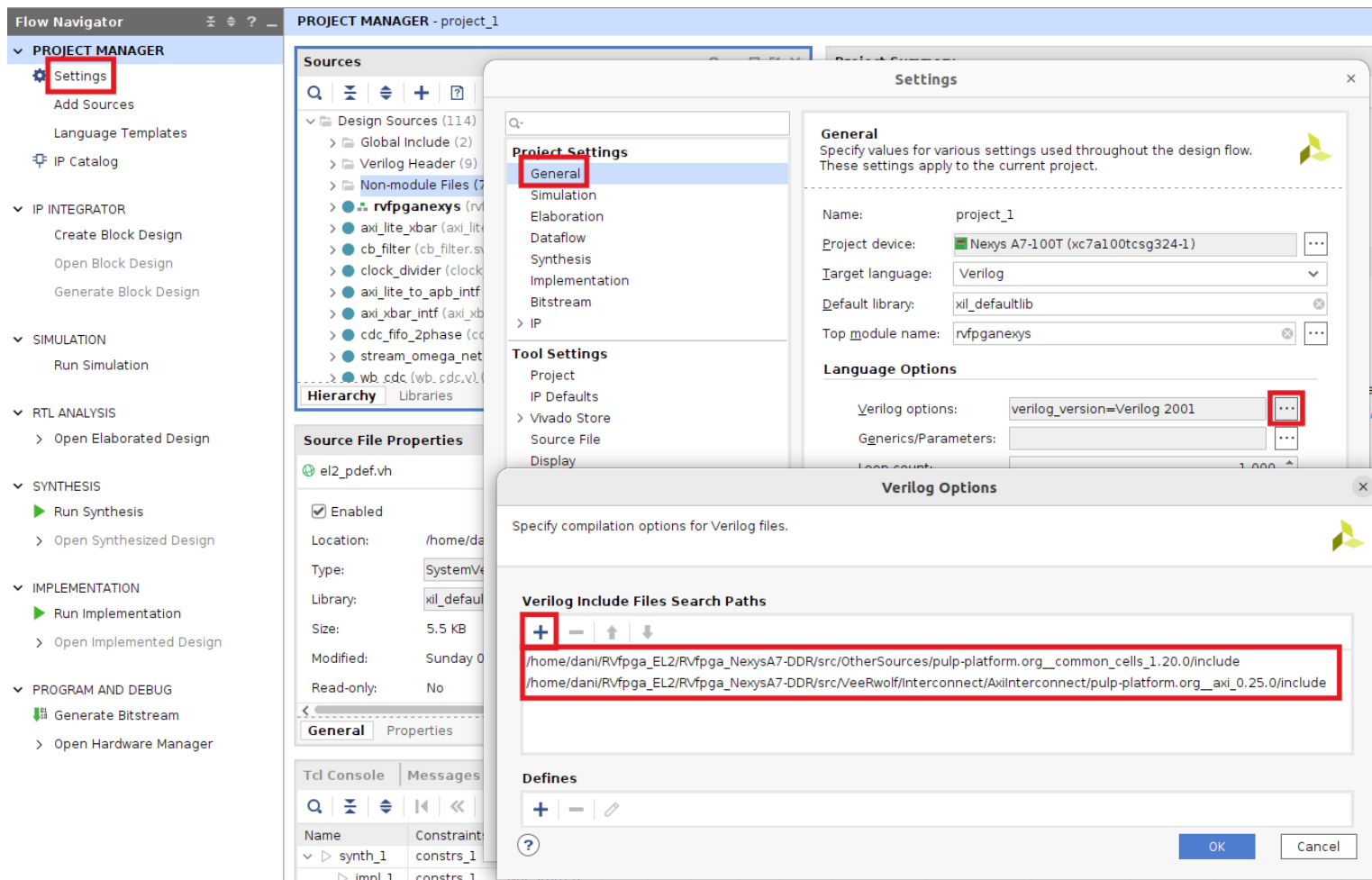


Figure 13. Include folders for the Pulp Platform

Add -sweep: Finally, add the `-sweep` option to the **Implementation – Opt Design**. See Figure 14.

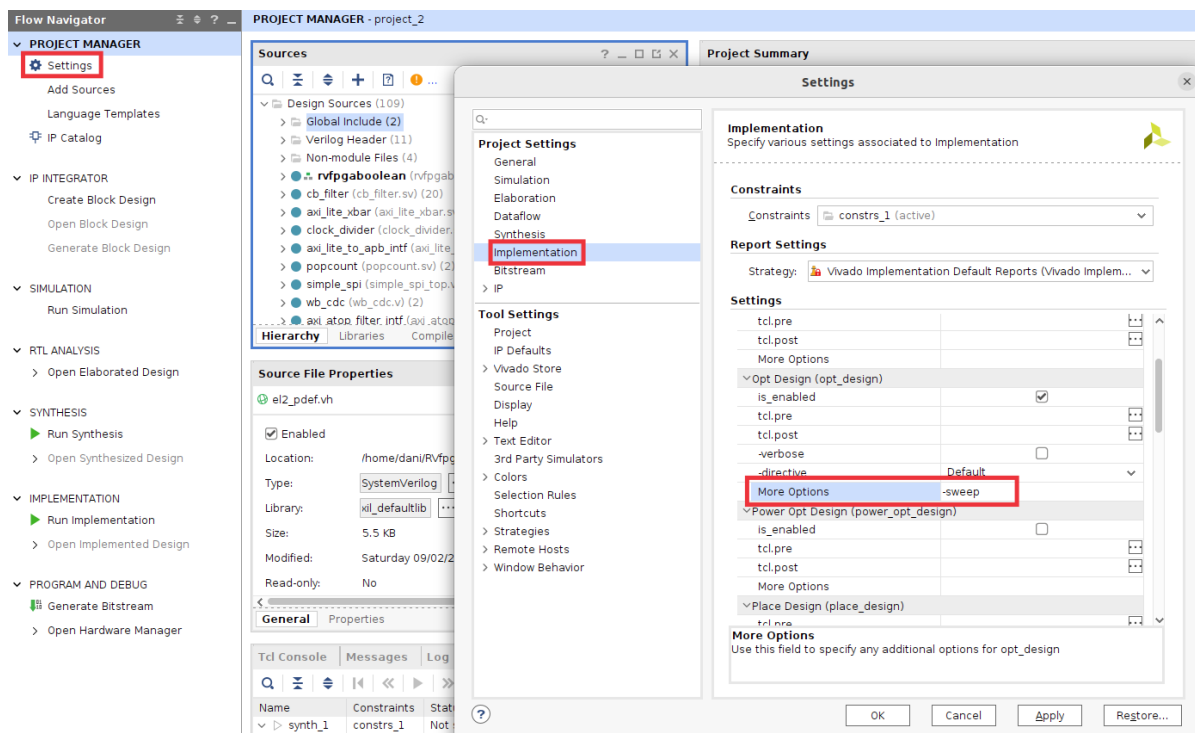


Figure 14. Add the –sweep option

Step 6. Generate Bitstream

Now Click on Flow → Generate Bitstream as shown in Figure 15. This process typically takes around 20 minutes, depending on the speed of your computer.

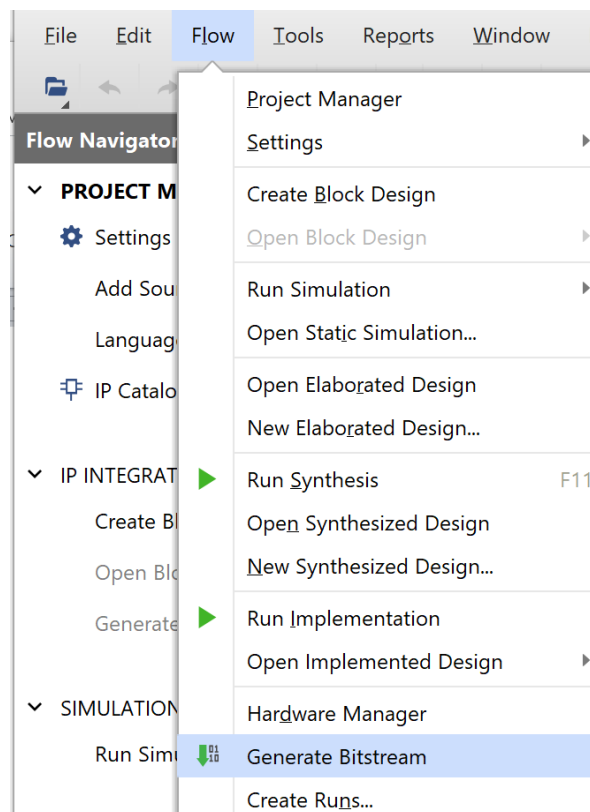


Figure 15. Generate Bitstream

After the bitstream has been generated a window will pop up as shown in Figure 16. Click on the **X** button in the top-right corner to close the window.

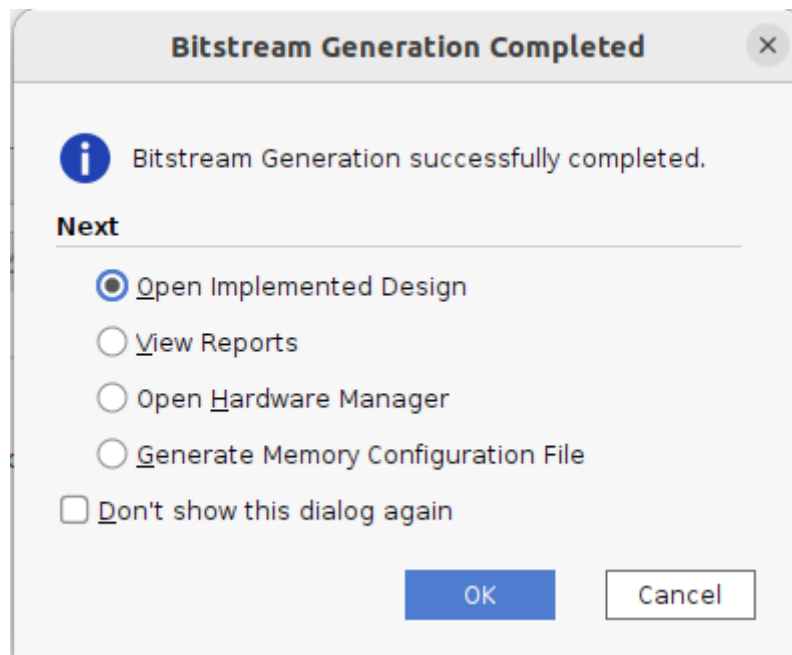


Figure 16. Bitstream Generation Completed

Now that you have built the RVfpgaEL2-Nexys VideoA7 system yourself, you will be able to rebuild RVfpgaEL2-Nexys VideoA7 after you make modifications to it in Labs 6-9. You can now also use the RVfpgaEL2-Nexys VideoA7 system you just built to download and run programs on it.

IMPORTANT: You can find the bitstream just generated by Vivado in folder:
--

<code>[RVfpgaEL2Nexys VideoNoDDRPath]/Labs/Lab05/project_1/project_1.runs/impl_1</code>

It is recommended that you use Catapult SDK to download RVfpgaEL2-Nexys VideoA7 onto the Nexys Video Video board, as described in Labs 1-4 and in the RVfpgaEL2 Getting Started Guide (GSG) in detail. As also described in the GSG and Labs 1-4, after downloading the RVfpgaEL2-Nexys VideoA7 system onto the FPGA on the Nexys Video Video board, you will use Catapult SDK to download and run/debug programs on RVfpgaEL2-Nexys VideoA7.